Università degli Studi di Parma
Dipartimento di Ingegneria e Architettura

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Cybersecurity
Exam 11/6/2025

1) Consider a message m encrypted with symmetric algorithm $E_K()$ and a key K obtaining the ciphertext $c=E_K(m)$. What do you need for carrying out a brute force attack?

- ☐ The ciphertext c and the encryption algorithm E()
- ☐ The ciphertext c, the encryption algorithm E(), and the key K
- ☐ The ciphertext c, the decryption algorithm D(.), and being able to recognize a valid plaintext m
- ☐ The ciphertext c, the encryption algorithm E(), and being able to recognize a valid plaintext m
- ☐ The ciphertext c and the decryption algorithm D()

2) Why a block cipher with block size equal to 16 bit cannot be considered secure?

- ☐ because it is easy to perform a brute force attack against the secret key
- ☐ because the encryption and decryption functions would be too fast to compute
- ☐ because it is possible to decrypt any ciphertext, without knowing the key, after obtaining $2^{\wedge 16}$ different $\{m_i,c_i\}$ pairs.
- ☐ because it is not possible to securely encrypt messages that are longer than 16 bits.

3) DSA is:

- ☐ A symmetric block cipher algorithm
- ☐ An asymmetric block cipher algorithm
- ☐ A hash algorithm
- ☐ A digital signature algorithm

4) Consider the following three padding algorithms for extending the length of a message to a multiple of N bytes (e.g. N=64). Which of the three algorithms is suitable for using with a hash function?

- ☐ append to the message a bit '1', then append as many bits '0' as needed to reach a multiple of N bytes.
- ☐ append to the message random bytes until the total length (in bytes) becomes a multiple of N.
- ☐ append to the message random bytes until the total length (in bytes) becomes a multiple on N – 1; append one byte encoding the number of padding bytes that have been added.

5) What do you need in order to verify the validity of a digital certificate?

- ☐ the private key of the CA that signed the given certificate
- ☐ the certificate of the CA that signed the given certificate
- ☐ your own certificate

6) Which of the following fields is NOT included within a X.509 certificate?

- ☐ the *private key* of the subject owner of the certificate
- ☐ the *subject* owner of the certificate
- ☐ the certificate *expiration date*
- ☐ the *issuer CA*, that is the CA that issued the certificate

7) What is usually used for protecting the TLS handshake between a client and a server?

- ☐ the certificate of the server
- ☐ IPSec encapsulation
- ☐ a remote authentication server
- ☐ a remote KDC (Key Distribution Center) server

8) What is a buffer overflow attack?

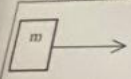
- ☐ an attack that exploits the vulnerability of a web app by trying to interfere with the queries that the app makes to its database
- ☐ an attack that exploits the vulnerability of a program that does not check the length of received or read data
- ☐ an attack that exploits the vulnerability of a LAN network switch that fails to process all received packets
- ☐ an attack that exploits the vulnerability of a firewall that does not maintain state information about previously processed packets

9) What is the meaning of the expression $a \equiv b \pmod n$? Write the mathematical relation between a and b.

10) Let us consider a message $m=M1\|M2\|M3\|M4$, and suppose to encrypt it by means of a block cipher $E_K()$ in CBC mode (the block size of $E_K()$ is equal to the size of the blocks Mi), with $iv=IV0$, obtaining the ciphertext $c=C1\|C2\|C3\|C4$. The c and iv are stored on a medium.
If an attacker modifies the iv and the ciphertext on the medium by writing iv'=IV1 and c'= $C1\|C3\|C3\|C4$, which will be the corresponding plaintext message $m'=M'1\|M'2\|M'3\|M'4$ obtained by "erroneously" decrypting the ciphertext c' with iv'? Show the blocks $M'i$ as function of $IV1$, Mj, and Cj, with $j=1..4$.

11) You captured a ciphertext $c=2$, encrypted using a RSA public key $K^+=<e,n>=<5,65>$. Try to discover the private key $K^-=<d,p,q>$ and decrypt c. (Hint: d is greater than 20).

12) We want to store a large message m (e.g. a file) onto an insecure public storage system, by guaranteeing both the confidentiality and the integrity/authenticity of the data m. Let's suppose to have a private/public key pair K^- and K^+, and to have the following cryptographic algorithms: RSA, AES, SHA2. Please indicate a possible functional scheme that can be used for such a purpose, and the resulting data that will be actually stored.



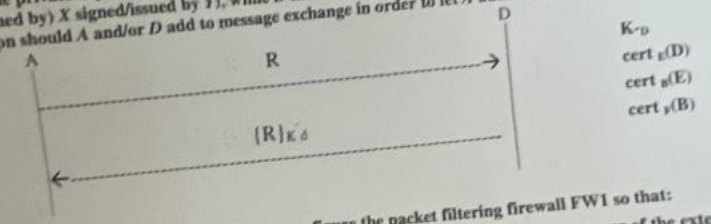
Stored data =

13) Show a possible challenge-response authentication scheme that can be used by Alice to authenticate Bob, based on a MAC function and a shared secret K_{AB}.

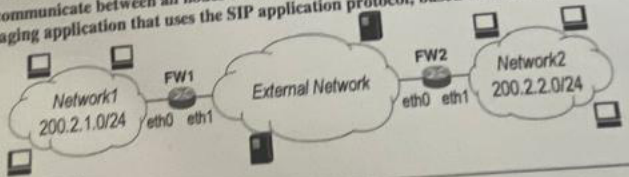


14) Show an example of successful Man-in-the-middle attack against basic Diffie-Hellman exchange between two entities A and B.

15) Let's consider the authentication scheme in figure where *A* wants to authenticate *D*. Consider that *R* is a random value and K^-_D is the private key of *D*). A has her own private key K^-_A, $cert_C(A)$ and $cert_C(B)$ (where $cert_Y(X)$ is a certificate of (owned by) *X* signed/issued by *Y*), while *D* has his own private key K^-_D, $cert_E(D)$, $cert_B(E)$, $cert_F(B)$. Which information should *A* and/or *D* add to message exchange in order to let *A* authenticate *D*?



16) Considering the following network topology, please configure the packet filtering firewall FW1 so that:
i) from network 1 it is possible to establish client-server HTTP communications to any web server of the external network (TCP server port 80);
ii) it is possible communicate between all nodes of networks 1 and 2 (client→server and server←client), by means of an Instant Messaging application that uses the SIP application protocol, based on UDP and server port 5060.



FORWARD		matching						action
in_int	out_int	src_addr	dest_addr	proto	src_port	dest_port	conn_state (NEW / ESTABLISHED)	ACCEPT / DROP